Turner Silverthorne

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Contact

University of Toronto
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Research Interests

Mechanistic models of circadian and neural systems, Circular statistics, Optimal experimental design, Scientific computing, Analysis of dynamical systems.

Education

University of Toronto

Ph.D. in Mathematics — Expected May 2027
Dissertation title: *Separation of timescales in mathematical chronobiology*
Advisors: Adam Stinchcombe and Arturas Petronis

M.Sc. in Mathematics — 2022
Thesis title: *A mathematical model of promoter methylation in the circadian clock*
Advisors: Adam Stinchcombe and Arturas Petronis

University of Waterloo

B.Sc. in Mathematical Physics, Minor in Pure Mathematics — 2020

Awards

Queen Elizabeth II Graduate Scholarships in Science and Technology	2026–2027
NSERC Canada Graduate Scholarship	2023–2026
Blyth Fellowship, University of Toronto (declined due to NSERC)	2023
Research Excellence Award, Canadian Society for Chronobiology	2023
Ontario Graduate Scholarship, Government of Ontario	2022
Margaret Isobel Elliott Graduate Scholarship	2022
NSERC Undergraduate Student Research Award	2019
NSERC Undergraduate Student Research Award	2017
Undergraduate Research Internship, University of Waterloo	2017
Undergraduate Research Internship, University of Waterloo	2016
President's Scholarship, University of Waterloo	2015

Research Experience

Research assistant, University of Vilnius (Life Sciences Center)	since 2022
Research trainee, Centre for Addiction and Mental Health (Neuroscience)	since 2020
Co-op student, Centre for Addiction and Mental Health (Neuroscience)	2018
Co-op student, University of Waterloo (Pure Mathematics)	2019
Co-op student, Institute for Quantum Computing (Pure Mathematics)	2017
Co-op student, Institute for Quantum Computing (Engineering)	2016

Publications

1. T Silverthorne, M Carlucci, A Petronis, AR Stinchcombe, *Optimization of experimental designs for biological rhythm discovery*. PLOS Computational Biology, **21**(11), e1013662 (2025).
2. [†]T Silverthorne and B Webster. *Gelfand-Tsetlin modules: canonicity and calculations*. Algebras and Representation Theory, **27**(2), 1405–1455 (2024).
3. [†]C Paddock, V Russo, T Silverthorne, W Slofstra. *Arkhipov's theorem, graph minors, and linear system nonlocal games*. Algebraic Combinatorics, **6**(4), 1119–1162 (2023).
4. T Silverthorne, ES Oh, AR Stinchcombe. *Promoter methylation in a mixed feedback loop circadian clock model*. Physical Review E, **105**(3), 034411 (2022).

([†]authorship in this field is alphabetical)

Preprints

1. M Carlucci[*], ES Oh[*], T Silverthorne[*], AR Stinchcombe, M Zelnys, A Petronis. *The chronology of developing cells: are epigenomic and transcriptomic oscillations linked to their linear trajectories?*. Preprint bioRxiv:2026.04.27.720664, 2026.

([*]these authors contributed equally)

Conferences and Talks

Invited talks

- [‡]Nonlinear dynamics in Biology, Canadian Mathematical Society Winter Meeting, Montreal 2026

Contributed talks

- Trainee Symposium, Canadian Society for Chronobiology Biennial Meeting, Montreal 2025
- Canadian Society for Chronobiology Biennial Meeting, Montreal 2025
- SRBR Biennial Meeting, San Juan, Puerto Rico 2024
- AMMCS Biennial Meeting, Wilfrid Laurier University 2023
- Canadian Society for Chronobiology Biennial Meeting, University of Guelph 2023

Posters

- CAIMS Annual Meeting (Virtual) 2021
- Canadian Mathematics Society Summer Meeting (Virtual) 2021
- Canadian Association of Physicists Congress, Queen's University 2017

Other talks

- Bird's Eye Conference 2025 & 2026
- Toronto Applied Math Learning Seminar. 2022–2026
- Math Graduate Student Seminar, University of Toronto. 2021 & 2026
- Physics of Information Lab, University of Waterloo. 2019

([†] confirmed, upcoming)

Professional Activities

Reviewer for Frontiers in Photobiology
Reviewer for Biometrical Journal

Committee Membership

Bird's Eye Conference, University of Toronto 2023–2024
Director (2024), Co-organizer (2023)

- Organized a two-day graduate conference for 60 attendees and 30 speakers, focusing on accessible communication between different fields of mathematics.
- Incorporated mentorship in the conference design, so that speakers would receive feedback and opportunities to improve the clarity of their presentations.
- Built a long-term framework for the event, including a mission statement and succession plan, which has allowed the conference to continue under new leadership.

Math Department Diversity and Equity Committee, University of Toronto. 2021–2022
Graduate Student Representative

- I served as a representative of the math graduate student community and contributed to a revised welcome package for graduate students emphasizing our community values, advice from senior students, and resources to promote diversity in the community.

Outreach

Graduate student mentor, University of Toronto since 2023

- Mentored new graduate students in the math department.

Math mentor, Math Outreach Office, University of Toronto 2021

- Guided two high school students on independent study projects.

Workshop Facilitator, FemPhys, University of Waterloo 2019–2020

- Developed and presented three interactive workshops on tools and strategies for undergraduate research.
- Provided opportunities for students with various backgrounds to discuss and learn from one another's research experiences.

Mentor, Let's Talk Science, University of Waterloo 2016–2018

- Conducted weekly mentoring sessions with a gifted middle school student focused on university-level topics in mathematics and physics.
- Presented an original online physics demonstration to a highschool in Nunavut.

TA Experience

University of Toronto

Ordinary Differential Equations for Engineers (MAT 292)	Fall 2026
Mathematical Modelling (APM 348)	Winter 2021
Calculus I (MAT 135)	Fall 2020

Graduate Coursework

University of Toronto

- Linear algebra and optimization
- Partial differential equations (I & II)
- Real analysis (I & II)
- Introduction to nonlinear evolution PDEs

- Numerical methods for mathematical modelling
- Cellular and molecular biophysics
- Foundational Discoveries in Genome Biology and Bioinformatics
- Deep learning: theory and data science

University of Waterloo
- Introduction to Lie groups and Lie algebras
- Introduction to General Relativity for Cosmology
- Applied Stochastic Processes

References

Adam Stinchcombe

Associate Professor, Department of Mathematics
University of Toronto
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Arturas Petronis

Professor, Departments of Psychiatry and Pharmacology & Toxicology
University of Toronto

Senior Scientist and Head, Krembil Family Epigenetics Laboratory
Centre for Addiction and Mental Health (CAMH)

Professor, Vilnius University
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Andreas Hilfinger

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Mary Pugh

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